

Received 25 June 2025, accepted 10 July 2025, date of publication 16 July 2025, date of current version 25 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3589653

RESEARCH ARTICLE

Internet Exchange Points IXPs: The Backbone of Efficient 5G Data Routing

OSMAN DIRIYE HUSSEIN^{ID1} AND ADAM MUHUDIN^{ID2}

¹Telecommunication Engineering Department, Faculty of Engineering, SIMAD University, Mogadishu 09010, Somalia

²Computer Science Department, Faculty of Computing, SIMAD University, Mogadishu 09010, Somalia

Corresponding author: Osman Diriye Hussein (osman@simad.edu.so)

ABSTRACT The accelerated deployment of 5G networks is bringing about a new age of high-speed connectivity, low-latency services, and large-scale device interconnectivity. The Internet Exchange Point (IXP) is at the forefront of this transformation and is essential to facilitate smooth data exchange among network operators, content providers, and end-users. IXPs are vital in lowering latency, optimizing bandwidth utilization, and enabling best possible data routing, all of which are essential to address the performance requirements of the 5G technology. Current IXP architectures, however, must be modified to support the explosive growth in data traffic, complex routing demands, and ultra-low latency needs of 5G applications such as autonomous vehicles, IoT, and virtual reality. This article proposes a next-generation IXP architecture optimized for 5G data routing. The architecture dramatically improves scalability, performance, security, and cost-efficiency by bringing together Software-Defined Networking (SDN), Network Function Virtualization (NFV), and edge computing to enable dynamic and localized traffic management. In addition, sophisticated security mechanisms, such as DDoS mitigation and Zero Trust models, are embedded in this architecture. The main findings are that the performance is improved and latency decreased with direct peering and optimized resource allocation. Despite implementation challenges, such as significant upfront investment and integration with existing systems, the architecture offers a future-proof and scalable solution. With Internet Exchange Points (IXPs) becoming ever more critical to the 5G ecosystem, their modernization is key to unlocking the future of global connectivity.

INDEX TERMS Internet exchange points (IXPs), 5G networks, low-latency routing, software-defined networking (SDN), edge computing, data routing.

I. INTRODUCTION

In this modern digital era, the rapid progress of technology fundamentally changed how information is communicated, processed, and used throughout the world. Among the technology strides in this respect, fifth-generation (5G) networks are at the forefront, setting new speed records, ensuring ultra-reliable low-latency communication, and smoothly connecting a vast number of devices. This transformative technology promises to be the bedrock on which a plethora of innovations in a myriad of fields—self-driving cars, smart cities, advanced healthcare, et cetera—rests. But with 5G networks now gaining inroads, the infrastructure that makes this

seamless connectivity possible has come into the limelight: most notably, Internet Exchange Points, or IXPs [1]. Often referred to as the backbone of efficient data exchange on the internet, IXPs are critical nodes where ISPs, CDNs, and other network operators interconnect. This brings the biggest benefit: by facilitating the direct data exchange between networks, IXPs reduce latency and improve bandwidth efficiency and the paths data takes to get to its destination. In a 5G context—where real-time communication and massive data throughput are taking precedence—IxPs will have grown in relevance. The deployment of 5G networks ushers in new paradigms for the ways that data moves within and between networks. Unlike its predecessors, 5G is not just about fast internet speeds; it's an ecosystem of connected devices where each will generate and consume data on a massive scale.

The associate editor coordinating the review of this manuscript and approving it for publication was Muhammad Khalil Afzal^{ID}.

This exponential rise in data traffic needs an efficient and scalable routing framework—that is where IXPs come into play. IXPs, by allowing localized traffic exchange, take off some load from the upstream transit providers and improve general network efficiency; hence, they are one of the critical elements in the successful deployment of 5G [2].

However, there is one very important determinant for 5G network performance, which is related to the geographical distribution of IXPs. The promise of 5G to offer ultra-low latency services makes proximity to IXPs a very important characteristic. A properly distributed network of IXPs sends data on the most efficient path to reduce latency and enhance users' satisfaction with the network. This becomes all the more important in applications such as augmented reality, virtual reality, and especially autonomous systems, where delays measured in a few milliseconds can have enormous consequences. Another very important feature of IXPs with relation to 5G technology lies in the edge computing area. With the coming of 5G mobile network technology, the fifth generation, through localized data processing, will be able to meet the requirements of applications that are latency-sensitive [3]. IXPs will then be part of the basic nodes in interconnecting edge data centers. Edge computing, together with IXP, can thus assure faster processing and transmission of data, further harmonizing the operation between smart systems and IoTs. Increasing emphasis on data sovereignty and regional traffic management of the internet further underscores the role of IXPs.

With 5G, the amount of data created has reached unprecedented levels. That caused countries and regions to focus on keeping their data within their borders for better security, privacy, and governance. A key piece in reaching that goal, IXPs help exchange traffic locally and reduce reliance on international transit paths. While they are among the most basic parts in integrating the 5G ecosystem, challenges still abound. Uneven global distribution of IXPs, unequal development of network infrastructure at various levels, and regulatory constraints may impede data routing optimization [2]. Such challenges can only be responded to by the joint efforts of governments, industry stakeholders, and technology developers, to make sure that everybody gets equal access to IXPs and their benefits. The article describes how 5G networks and IXPs interlink with each other to improve efficiency in data routing, supporting edge computing and the solution of latency issues. With case studies and current trends, it is clear that IXPs are playing a fundamental role in this transformation triggered by 5G.

This, therefore, becomes a very key area of research, since IXPs are now being considered not only as enablers but as very crucial catalysts in fully realizing the potential of 5G networks and, more importantly, in shaping the future of connectivity and innovation. Increased data traffic and energy consumption in 5G networks come with environmental implications that can't be overlooked. IXP, by optimizing data exchange and reducing superfluous traffic, contributes to a

more sustainable Internet infrastructure. The potential for the reduction of duplicate data pathways brings with it increased efficiency but also lowers the general energy consumption of network operations across the globe. That makes IXP one of the most important entities building a greener and much more sustainable digital ecosystem [4].

II. LITERATURE REVIEW

The development of 5G technology has motivated extensive research in different areas, from network architecture to data routing mechanisms and even interconnection infrastructures such as IXPs. These works point out the difficulties and opportunities in the deployment of 5G networks with efficiency, reliability, and scalability.

A. 5G TECHNOLOGIES

The basic building blocks of a 5G network were discussed in a number of works, detailing how they differ from the previous generations of mobile communications. Research on massive machine-type communication (mMTC), ultra-reliable low-latency communication (URLLC), and enhanced mobile broadband (eMBB) has concluded that these are the main enablers of the 5G ecosystem. Academia has researched each of these capabilities applied within different fields [5]. The first, mMTC, will be indispensable for certain specific IoT applications, such as smart cities and industrial automation. URLLC, with its stringent latency requirements, has been studied in contexts like autonomous vehicles and telemedicine. Similarly, the high-bandwidth capabilities of eMBB are envisioned to be a key driver for applications such as high-definition video streaming and virtual reality. However, these studies have constantly pointed out the challenges in scaling 5G. The densification of network architecture—its logical consequence for reaching the set performance benchmarks of 5G—poses a particular challenge in data routing and interconnection. Exacerbating this challenge further is the exponential rise in data traffic that needs efficient mechanisms in place to manage and direct it accordingly.

B. IXPS AS EFFICIENT INTERCONNECTION ENABLERS

The role of IXPs in modern network infrastructures has been a recurrent theme in research. IXPs allow the local exchange of traffic, thus decreasing dependence on upstream providers and improving latency and bandwidth utilization. Additional studies have stressed their role in maintaining the stability of the internet through the provision of backup routes and improving the reliability of networks. Key contributions in this field include the analysis of how IXPs work as a localized traffic exchange hub, thereby reducing the latency in long-haul data transmissions. They pointed out that ISPs and content providers achieve cost savings through leveraging IXPs, which reduce transit costs and allow for more efficient use of network resources [6].

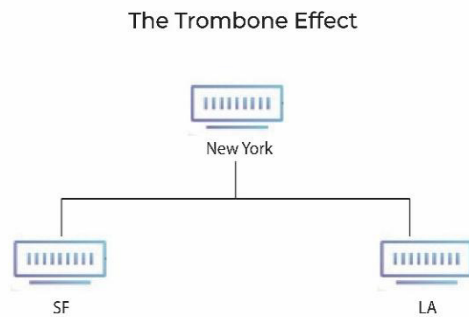


FIGURE 1. The trombone effect in network routing.

C. IXPS IN THE CONTEXT OF 5G

The role that IXPs will play in 5G networks has been addressed in both academic and industry literature. Given the promise of unprecedented volumes of data by the 5G networks, IXPs have increasingly become recognized as central to dealing with this increase. In other words, IXPs are a means of reducing latency and guaranteeing high throughput—features inherent in 5G applications—since they make possible direct interconnection among networks. Much research attention has been directed at the question of interoperability between IXPs and the requirements that 5G services demand. For example, this article investigates how IXPs can meet the low-latency requirements of recently invented real-time applications, such as augmented reality, virtual reality, and telepresence. Furthermore, the role of IXPs in supporting traffic exchange for M2M communications, especially those related to IoT devices, is highlighted [7].

The massive traffic streams, resulting from the proliferation of IoT in 5G environments, require efficient interconnection in order to avoid bottlenecks. IXPs ease these pains by providing localized routing solutions that not only improve speed but also lower operational costs for service providers. Moreover, IXPs play a very important role in load balancing and congestion control, which are two of the most critical aspects related to stability in the 5G network, especially at peak times. In this respect, IXPs prevent service quality disruptions of applications that demand ultra-reliable low-latency communications (URLLC) by easily allowing traffic to be transferred and reallocated. Such functionality makes IXPs crucial in realizing the full potential of 5G's advanced communication paradigms.

D. EDGE COMPUTING AND IXP

Most of the related works converge on the convergence of edge computing and IXP. Edge computing, as an approach that moves data processing closer to its source, has been identified as a critical enabler of low-latency and high-efficiency data transmission. Researchers have noted that IXPs act as strategic hubs for integrating edge data centers, thus enabling seamless interaction between edge and core

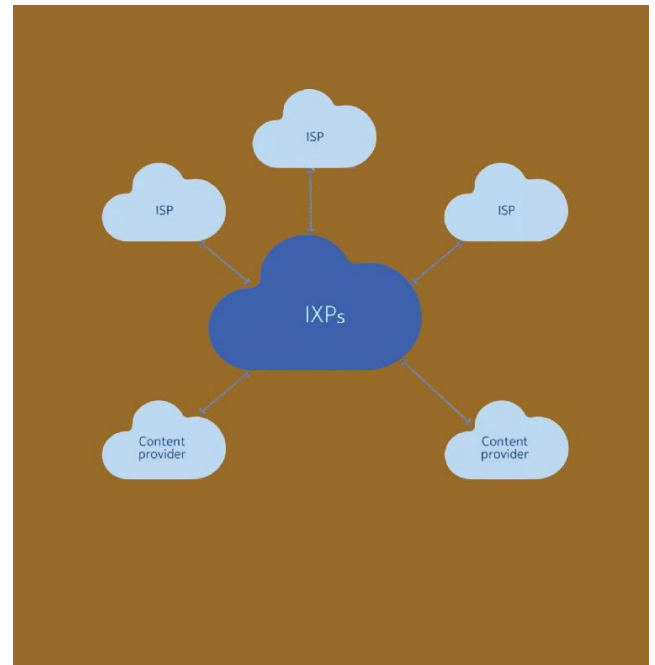


FIGURE 2. IXPs manage traffic between ISPs and content providers.

networks. Specifically, various research has been conducted on how IXPs improve the performance of latency-sensitive applications by localizing traffic exchange [8]. This is very important for applications like smart grid monitoring, autonomous vehicles, and remote healthcare, which, even with small delays, might lead to serious consequences.

E. GEOGRAPHICAL DISPARITIES IN IXP DISTRIBUTION

A recurring theme in research is the uneven global distribution of IXPs. A number of works have also documented the concentration of the IXPs in developed regions that enjoy better infrastructure and investment. Poorly served, developing regions usually experience a lack of IXP that leads to higher reliance on international transit and higher latency. Efforts toward the addressing of these disparities are also well-documented in literature. The Af-IX, and other regional associations have engaged in the development of IXPs to enhance better local connectivity that would respond to emerging technologies such as 5G. The previous discussion provides critical context to the challenges faced and the strategies that need to be adopted to make the global network infrastructure more equitable [9].

F. REGULATORY AND GOVERNANCE CONSIDERATIONS

Other related works also discussed the regulatory and governance aspects of the IXP. Given that IXPs are becoming central in 5G deployment, their governance structures and regulatory frameworks gain prominence. There has been thought over how IXP positioning can be made in tune with national data sovereignty laws dictating that the data created within a country stays within its borders. The role of

IXPs in strengthening cybersecurity has also been examined. Acting as a controlled point of interconnection, the IXPs provide better management and monitoring of traffic; thus, there is little scope for cyberattacks to happen. This is quite relevant to 5G networks, wherein the high volume and complexity of data traffic can make it susceptible to security threats [10].

G. EMERGING TRENDS: SOFTWARE-DEFINED NETWORKING (SDN) AND NETWORK FUNCTION VIRTUALIZATION (NFV)

Advanced technologies like SDN and NFV are being widely adopted in IXPs, enabling more dynamic and flexible network resource management-capabilities at the heart of fulfilling demanding 5G networks. Researchers have explored exactly how SDN and NFV will facilitate IXPs to scale, adapt to shifting traffic patterns, and support a wide diversity of service requirements. These works show a direction toward the development of an IXP that can work in future network architectures [11].

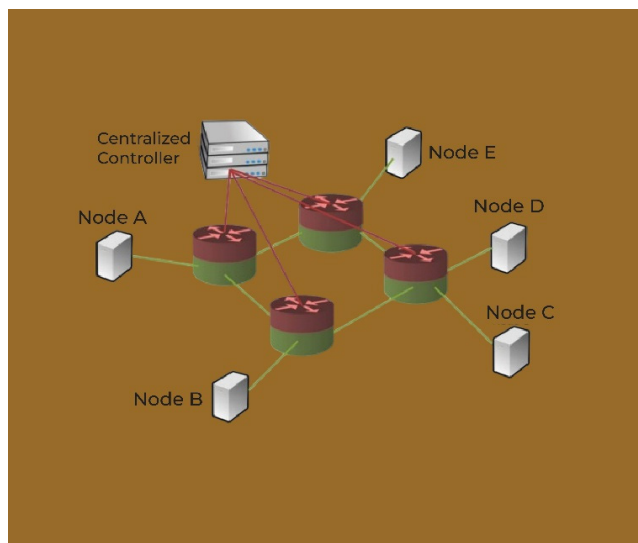


FIGURE 3. Software-defined networks and network function virtualization.

H. ENVIRONMENTAL SUSTAINABILITY

Some research has focused on the environmental implications of IXPs and their role in fostering sustainable internet infrastructure. A few studies analyzed how IXPs could reduce the energy footprint of data transmission through path optimization of routing and by minimizing redundant traffic. This supports global efforts toward green ICT solutions to reduce carbon emissions related to digital technologies. Now, considering this in relation to 5G, IXP integration and sustainable practices become even more critical [12]. Researchers have discussed how IXPs can host energy-efficient network architectures and hence contribute to the overall sustainability of the 5G ecosystem.

III. PROPOSED ARCHITECTURE FOR IXPS AND 5G DATA ROUTING

The current digital age is characterized by unparalleled connectivity and significant flow of data. The emergence of technologies such as 5G has made the requirement for high-speed, efficient, and flexible networks all the more critical. The interconnection point (IXP) is one of the most crucial elements that ensures continuity of connectivity and efficient data routing. The IXP enables direct traffic exchange between Internet Service Providers (ISPs), content providers, and enterprises. An efficient IXP architecture cannot be undermined with increasing complexity related to the Internet and with the dawn of 5G networks. Following is the detailed architecture for the IXPs that will form the basic framework for efficient 5G data routing. The proposed architecture attempts to enhance the performance, reduce the latency, and ensure the scalability, therefore fitting the next-generation network requirements such as 5G [13].

A. IXP ARCHITECTURE DESIGN

The IXP architecture is designed with a focus on scalability and low-latency data routing. The architecture integrates multi-layer switching technologies to optimize data traffic flow.

B. EDGE SWITCHING LAYER

This layer is accountable for the initial traffic aggregation from multiple Internet Service Providers (ISPs) and local networks. Packet filtering, QoS management, and traffic prioritization are carried out by this layer.

C. CORE SWITCHING LAYER

The core layer handles the high-speed switching of data between edge layers. It is designed to minimize latency and maximize throughput using Software-Defined Networking (SDN) technologies.

D. PEERING AND INTERCONNECT LAYER

At this layer, the data is exchanged between different ISPs and content providers through peering agreements. This facilitates faster and more direct routing paths.

IV. THE ROLE OF IXPPS IN 5G

The interconnection of multiple ISPs, content providers, and network operators, either bilaterally to exchange data traffic at these physical locations, is called the Internet Exchange Point (IXPs). Traditionally, IXPs enabled regional ISPs to communicate directly with each other, bypassing expensive long-haul transit service. With a 5G environment, it becomes even more important, in light of heavier data traffic volume, a heightened demand for reduction in latency, and routing efficiency. Some of the critical factors in delivering advanced services like autonomous vehicles, IoT, AR/VR, and more in a 5G environment include low latency and high throughput [14]. IXPs help to fulfill such demands by offering a platform where networks can exchange traffic

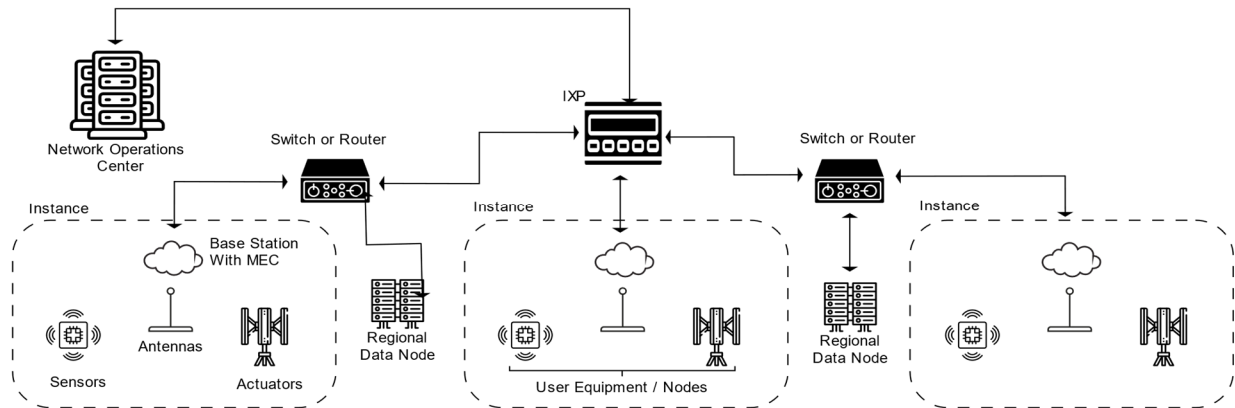


FIGURE 4. Architecture IXP and 5G for NOC.

without unnecessary detours, ensuring faster and more direct routing. As the scale and complexity of 5G networks increase, there will be a correspondingly greater need for IXP architectures to support such demands, ensuring the optimum flow of data and reduced congestion.

A. KEY OBJECTIVES OF THE PROPOSED IXP ARCHITECTURE

The proposed architecture of the Internet Exchange Point (IXP) for 5G data routing is driven by some key objectives.

1) SCALABILITY

The architecture of IXP will have to scale seamlessly, supporting increasing demand for high-speed data and a growing number of connected devices.

2) LOW LATENCY

An IXP must be designed in a way that the applications that require extremely low latency—augmented reality, virtual reality, and connected vehicles—face minimum delay in routing the data.

3) RESILIENCY AND RELIABILITY

The 5G service can only be delivered without disruption if the network resiliency is ensured. IXP architecture has to be highly available and fault-tolerant.

4) TRAFFIC OPTIMIZATION

Efficient routing mechanisms must be in place to minimize gratuitous data hops, avoiding bottlenecks to ensure optimal performance. As more and more devices rely on 5G, not to mention critical applications running on 5G networks, it will be necessary for the IXP to assure effective security measures.

5) COST EFFICIENCY

As traffic volumes rise, IXPs need to be designed to have the lowest possible operational costs by decreasing

dependency on third-party transit providers and efficiently using infrastructure.

B. COMPONENTS OF THE IXP ARCHITECTURE

The switching fabric resides at the heart of any Internet Exchange Point (IXP), enabling interconnection among multiple participants and, therefore, allowing data exchanges [15]. In order to be able to deliver high throughput and thereby support the enormous amounts of traffic that 5G networks will generate, switching fabrics must have the following attributes:

1) HIGH PERFORMANCE

It must support ultra-high-speed connectivity, say 100G or 400G, in a manner that guarantees smooth flowing data without congestion.

2) MODULAR

The switching fabric should be modular, allowing for easy expansion as the network grows.

3) REDUNDANT

For high availability, inherent redundancy such as dual switches and multiple data paths shall be designed in the switching fabric to avoid single points of failure.

4) LOW LATENCY

The design must reduce the number of hops and eliminate any unnecessary intermediaries to avoid latency in data delivery, as required by 5G use cases.

C. DISTRIBUTED ARCHITECTURE

The 5G network IXP covering vast geographical regions must take a distributed architecture. This will integrate numerous IXPs into a global fabric with efficient routing within regions. Distributed IXP network lowers latency by maintaining local exchange of traffic and averting long-distance routing.

Additionally, the approach provides redundancy, making it fault-tolerant [16].

1) LOCAL IXPS

These are small IXPs located near end-users in order to ensure low latency in exchanging traffic within the region.

2) REGIONAL IXPS

These IXPs act as an aggregation point for several local IXPs and have higher capacity, allowing interconnection among various regions.

3) GLOBAL IXPS

These are the top-level IXPs and act as the central hubs for intercontinental traffic exchange. The IXPs aggregate traffic from multiple regions and act as the backbone of the network.

D. PEERING ECOSYSTEM

One of the core benefits of IXPs is their peering ecosystem. Peering allows networks to exchange data directly, avoiding the need for costly transit providers [17]. The IXP should support multiple peering models in order to accommodate different types of participants:

1) PUBLIC PEERING

In this model, all the members share the same physical infrastructure and exchange traffic via a common switch. The model is ideal for small ISPs, content providers, and enterprises that want to interconnect with a large number of networks.

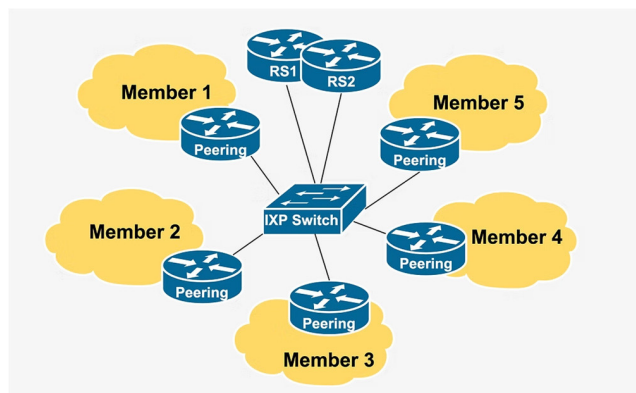


FIGURE 5. Public peering design system.

2) PRIVATE PEERING

This form of peering involves dedicated, direct interconnects between two participants. It would, therefore, be used mainly for high-volume or high-priority traffic in which reliability and performance are required.

The IXP should support dynamic peering agreements to allow participants to adjust their peering arrangements as network demands change.

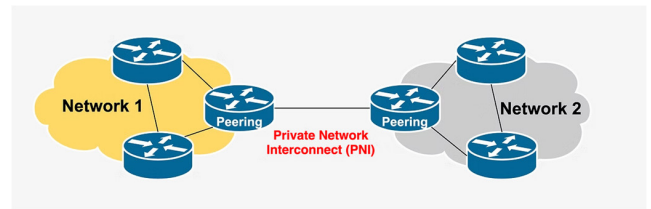


FIGURE 6. Private peering design system.

E. EDGE COMPUTING INTEGRATION

With the 5G focus on edge computing, there is a need for the IXP architecture to include edge data centers where content can be cached closer to end-users, reducing latency and improving user experience. Edge data centers integrated with IXPs keep data processing closer to its source, instead of being transmitted to distant centralized data centers [18].

1) EDGE DATA CENTERS

These centers shall have a direct connection to the IXP for the quick delivery of content to the users. They will also house local application servers, media contents, and even computational resources for IoT devices.

2) CONTENT CACHING

Caching at the edge reduces the need for retrieving data from distant sources, hence improving performance while reducing congestion at the core of the network.

3) LOW LATENCY PROCESSING

Edge computing can enable real-time processing of data, which is critical for applications such as autonomous vehicles, industrial IoT, and real-time video streaming.

F. SECURITY AND DDOS MITIGATION

Security is one of the most critical concerns for 5G networks, supporting a wide range of devices and services. The IXP must be designed to defend against cyber threats, especially Distributed Denial-of-Service (DDoS) attacks, which are the kinds of attacks that could flood network resources [19].

1) DDOS PROTECTION

IXPs should have the best-in-class DDoS mitigation technologies that can detect and mitigate large-scale attacks in real-time with the minimum possible disruption of services.

2) TRAFFIC SCRUBBING

A traffic scrubbing system would clean malicious or unwanted traffic before it gets into the network, protecting the integrity of both the IXP and connected networks.

3) SECURITY

The Internet Exchange Point (IXP) will follow an architecture that provides encrypted peering sessions to secure sensitive data as it is transmitted across the IXP.

G. AUTOMATION AND SDN INTEGRATION

Software-Defined Networking (SDN) provides the capability to manage the network from a centralized point, enabling the optimization of traffic flows and the making of dynamic changes based on events occurring within the network. The IXP will integrate SDN technologies into the design to enhance the management and operation of the network [20].

1) DYNAMIC TRAFFIC MANAGEMENT

SDN allows the IXP to change routes dynamically, considering factors like network congestion, traffic patterns, and others for optimal routing at all times.

2) AUTOMATED PROVISIONING

The provisioning of new connections should be automated, making the process easier for participants and reducing administrative overhead.

3) NFV

NFV gives much more flexibility because it virtualizes and dynamically deploys network functions—such as routing, firewalls, and load balancing—on demand.

H. MONITORING AND ANALYTICS

Real-time monitoring and analytics will be necessary for better understanding of the network performance and detection of any possible issues well in advance [21]. IXP shall implement monitoring systems that track network health, traffic volumes, and SLA compliance.

1) REAL-TIME MONITORING

Continual traffic flow, network health, and performance metric monitoring ensure the fast detection and resolution of issues.

2) TRAFFIC ANALYSIS

Analytics can be applied to gain insight into traffic patterns for use in routing optimization and planning capacity expansion.

3) PROACTIVE FAULT DETECTION

The automated fault-detection systems will predict failures or disruptions before they occur to ensure continuity of service.

The new architecture of IXP has been designed with the motive of meeting increased demands from modern networks in terms of scalability, low latency, and optimal performance for 5G data routing. Built using state-of-the-art technologies of edge computing, SDN, and AI, IXP is going to take up a central role in the efficient exchange of data across the global network. As the Internet expands, IXPs will remain agile in adding new features to facilitate future technologies such as quantum computing and 6G [18]. They will remain the backbone for efficient data routing for 5G and beyond through their infrastructure development and addition of cutting-edge best practices.

V. IMPLEMENTATION

The deployment of the proposed architecture for the Internet Exchange Point as it relates to routing 5G data is extremely complicated and multidimensional. In this context, meticulous planning, robust technical execution, and coordination with a broad array of stakeholders will be essential. This section delineates the most important steps and strategies concerning the deployment of the suggested architecture, from the establishment of the infrastructure to its ongoing enhancement.

A. INFRASTRUCTURE ESTABLISHMENT AND DEPLOYMENT

Physical and network infrastructure forms the cornerstone of any sound IXP design. This extends from site selection for the IXP to environmental preparation inside the data center, to connecting it with the respective ISPs, content providers, and cloud platforms.

B. IXP LOCATION AND SITE REQUIREMENTS SELECTION

Physical location is of the utmost importance in an IXP in terms of cost-effectiveness, reliability, and performance.

C. GEOGRAPHICAL PROXIMITY TO THE MAJOR NETWORKS

The location where the IXP is placed proximal to any big agglomeration of ISPs, Cloud Data Centers, and Enterprise hubs where the traffic exchange can take place in the best manner possible. It lowers latency when the IXP is proximal to major backbone networks.

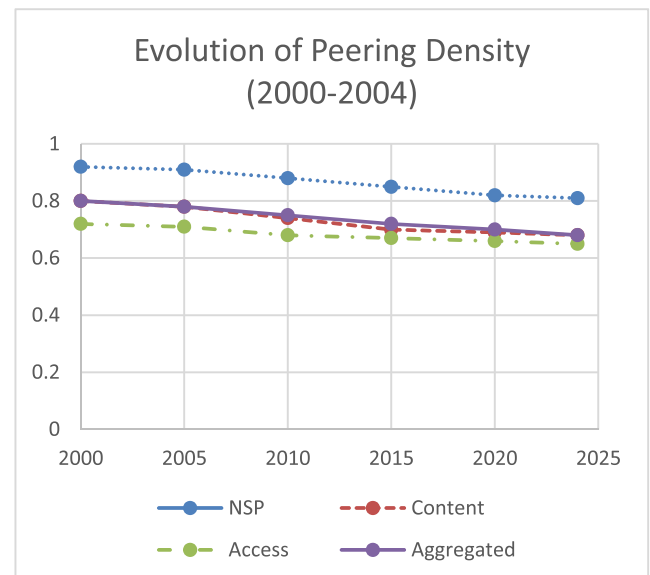


FIGURE 7. Geographical proximity of peer density for IXPs.

1) SECURITY AND PHYSICAL REDUNDANCY

The IXP should be housed in a facility with strong physical security—access control, surveillance, and fire-prevention

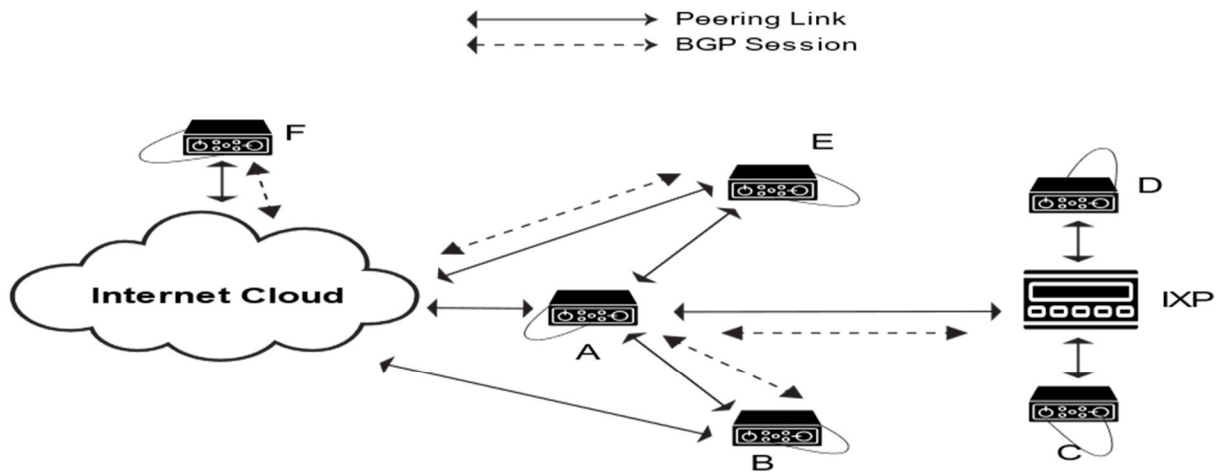


FIGURE 8. Cloud peering & BGP for IXP internet.

systems—that ensures the building’s infrastructure. Also, of utmost importance for high availability are power supply, cooling system, and network link redundancy.

2) NETWORK EQUIPMENT PROCUREMENT AND SETUP

To achieve the high-throughput, low-latency requirements of the system, the network equipment selection must be quite precise.

3) HIGH-SPEED SWITCHES

Inside the IXP switching architecture, there will be high-capacity, low-latency switches (100G, 400G) that are meant to efficiently process large amounts of data. Only equipment from leading manufacturers such as Cisco, Juniper, or Arista would be suitable due to their past track record of providing reliable, high-performance solutions.

4) REDUNDANT ROUTERS

Routers must be deployed for traffic flow control and distribution between networks. These must support both public and private peering and provide data routing on various routes on the network.

5) EDGE DEVICES AND SERVERS

Specialized devices like high-computing-power, low-latency servers should be installed for edge computing. These servers will run applications and services requiring immediate data processing and will be integrated into the IXP for direct local data exchange.

6) NETWORK TRAFFIC MANAGEMENT AND MONITORING TOOLS

Implement network monitoring and traffic analysis tools to provide real-time insight into the performance of the IXP. The routing could be further optimized, traffic patterns analyzed, and problems quickly troubleshooted by using network

performance monitoring software such as SolarWinds or NetFlow Analyzer.

D. ESTABLISHING CONNECTIVITY

Next in the chain of process is connecting it with the outside networks that are going to form a part of the IXP. Following are the actions that need to be initiated:

1) PEERING AGREEMENTS

Formal agreements among the ISPs, content providers, and cloud services are needed to ensure proper peering. Agreements will involve both public and private peering models based on the needs of the participants.

2) DIRECT LINKS TO CLOUD SERVICE PROVIDERS

Establish direct linkages with major cloud service providers so that the enterprises can have access to the cloud service without much latency, which is achieved using the service of AWS Direct Connect and Microsoft Azure ExpressRoute and Google Cloud Interconnect for exchange of traffic between cloud and on-premises.

3) MULTILATERAL PEERING

Every relationship is multilaterally peered, where several networks simultaneously exchange traffic via the IXP, interconnecting to each other in an ecosystem. This helps in ensuring that the flow of data is efficient, even when particular peers encounter network congestion or failures.

E. NETWORK REDUNDANCY AND FAILOVER MECHANISMS

Redundancy and failover mechanisms in network design will ensure that the IXP is available anytime and is fault-tolerant:

1) DUAL POWER SUPPLIES

The IXP should have duplicate power feeds. This ensures continuous operation if one of the feeding powers fails. UPS units, coupled with backup generators, also help prevent any downtime.

2) MULTIPLE NETWORK PATHS

The network should have multiples of paths to transmit data through and have automatic failovers in place in case of failures of one link; the traffic would automatically reroute through another path without impact.

3) LOAD BALANCING

Running the algorithms of load balancing on various network devices in order to prevent congestion on any single link by distributing the traffic evenly.

F. SDN INTEGRATION

Integration of SDN is very crucial for automation and optimization of network management. SDN provides IXP with the facility of centralized control over network infrastructure; hence, managing traffic flows, configuring routing policies, and optimizing network resources will be much easier. It acts as the brain in the functioning of the entire SDN; the controller that makes decisions intra-network. Currently popular ones are Open Daylight, ONOS, and Cisco's APIC. In a network setup, an SDN controller will need to be deployed to facilitate automation for routing traffic dynamically with respect to runtime network status conditions for preventing congestion, decreasing latency, or optimization.

1) FACILITATING NETWORK SLICING

Network slicing in 5G will enable an IXP to create virtual networks dedicated to specific applications. An SDN controller can dynamically allocate resources for different slices based on demand.

2) VIRTUALIZATION OF NETWORKS AND CONSOLIDATION OF RESOURCES

Network Functions Virtualization (NFV) is applied in creating virtual instances of the aforementioned networking functions, such as firewalls, load balancing, and traffic analysis.

3) ELASTIC RESOURCE ALLOCATION

The virtualization of network functions makes them dynamically adaptive to the varying traffic demand; hence, IXP infrastructure is constantly optimized and never over-provisioned.

4) BETTER COST EFFICIENCY

IXPs improve the overall efficiency of virtualized network functions and decrease capital expenditures by using standard hardware. The SDN controller is a centralized

decision point that orchestrates real-time network behavior across virtualized infrastructure. It is accountable for the dynamic provisioning and orchestrating of network slices to fulfill exacting 5G service demands, including enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), and massive Machine Type Communications (mMTC). By interfacing with NFV components, the controller allocates virtualized resources such as bandwidth, processing capacity, and routing paths based on service priority, traffic type, and latency sensitivity. It employs programmable APIs to inform edge nodes and core switches to enforce traffic engineering policies, enabling optimal performance and isolation between slices. Dynamic resource assignment enables the IXP to support various QoS demands while delivering agility and scalability in the face of evolving network conditions.

G. EDGE COMPUTING INTEGRATION

Edge Computing Integration of edge computing with the IXP is a must for 5G applications that require real-time processing of data, strategically deploy Edge servers within the IXP to reduce latency, they will host computing resources and services that need to process data near end users, key steps include edge data center placement, the placement of edge data centers that are geographically close to the IXP and can host applications requiring low-latency responses, local content caching implement content caching systems to keep frequently accessed content closer to the user to enhance performance while reducing the load on core data centers. Building Traffic Routing to Edge Servers With edge servers in place, the IXP has to facilitate traffic routing to those edge servers so that data is processed locally, this includes the following direct traffic to the Edge enforce protocols and routing policies that direct traffic to edge servers in order to minimize backhaul to distant, centralized data centers, Real-time processing in IXPs for data created by applications like IoT, autonomous vehicles, and video streaming must be done in edge servers themselves to support real-time latency-sensitive services.

H. SECURITY IMPLEMENTATION

Security for the IXP is of great concern, now more with upcoming 5G and associated vulnerabilities hence, strong security mechanisms are to be implemented Which will include but is not limited to DDoS Mitigation and Traffic Scrubbing the IXP should implement sophisticated.

DDoS detection systems to identify malicious traffic patterns and trigger the mitigation strategies in real time Traffic filtering and scrubbing systems should be installed where the incoming traffic is cleaned up by cleaning the traffic of malicious packets before reaching the core infrastructural elements of the IXP Rate limiting and Traffic Shaping. Deploy the rate limiting and traffic shaping mechanisms that ensure DDoS attacks do not cause resource overloading within the IXP Zero Trust Architecture.

The IXP should be designed to implement Zero Trust architecture to ensure that all sessions are authenticated and authorized to exchange data End-to-End Encryption. All peering sessions and data exchanges should be encrypted to ensure that data is secured during transmission Role-Based Access Control Establish RBAC policies that will prevent unauthorized access to the critical infrastructure elements, ensuring that only granted entities are allowed to make changes to the IXP configuration.

I. MONITORING AND OPTIMIZATION

The final implementation step involves constant monitoring, management, and optimization of the IXP architecture. This Real-Time Monitoring Tools Leverage network monitoring tools to provide real-time data on network performance, traffic patterns, and resource utilization:

1) TRAFFIC FLOW ANALYSIS

Utilize analytic tools-such as NetFlow or Flow-to monitor traffic flow and identify any bottlenecks or inefficiencies in the underlying network infrastructure.

2) FAULT DETECTION

Provide mechanisms for automatic fault detection that, by using some form of predictive analysis, automatically notify operators so proactive actions may be taken to avoid service disruption.

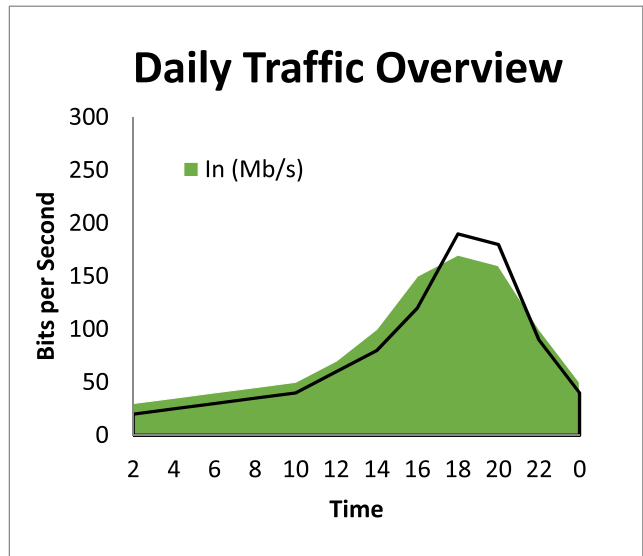


FIGURE 9. Daily traffic overview.

VI. ILLUSTRATIVE PERFORMANCE EVALUATION

Although this study does not include full-scale simulations, we provide an illustrative performance comparison based on realistic design assumptions. These charts highlight potential improvements in latency and throughput when using the proposed IXP architecture compared to traditional IXPs, under varying load scenarios.

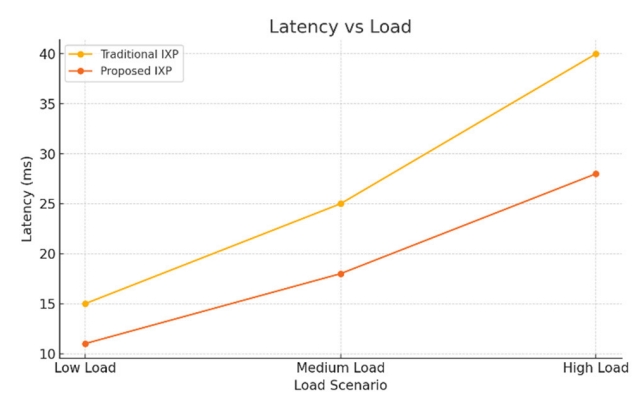


FIGURE 10. Comparison of average latency between traditional and proposed IXP under different traffic loads.

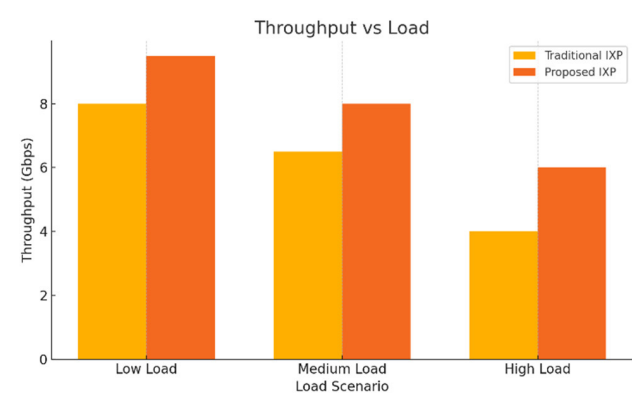


FIGURE 11. Variation of throughputs between conventional and proposed IXP architecture under assumed 5G traffic patterns.

VII. MIGRATION FROM LEGACY IXPS

The migration strategy is mapped onto a hybrid model integrating Software-Defined Networking (SDN) and Network Function Virtualization (NFV) elements onto the existing Internet Exchange Point (IXP) infrastructure. The integration enables service providers to introduce programmable control and virtualization in phases, thereby reducing operational disruptions. The process starts with the parallel deployment of SDN controllers with conventional routing equipment, thereby enabling policy-based traffic control and management. Subsequently, Network Function Virtualization (NFV) components facilitate the virtualization of essential core network functions such as load balancing, traffic filtering, and security policy enforcement. Compatibility modules provide bridges for allowing interaction between the legacy systems and the new architecture. Stepwise enhancement provides backward compatibility, cost management, and minimal risk while creating foundations for comprehensive modernization of the Internet Exchange Point (IXP). Legacy elements can be phased out as virtualized infrastructure assumes primary roles.

VIII. DISCUSSION

The proposed architecture for 5G Internet Exchange Points (IXPs) in data routing responds to the growing demands of current networks in terms of low latency, high throughput, scalability, security, and affordability.

A. BENEFITS

1) BETTER PERFORMANCE

Edge computing and low-latency routing support real-time applications in the form of IoT and autonomous vehicles.

2) SCALABILITY

The SDN/NFV-integrated modular architecture makes it easy to expand, handling dynamically increasing data traffic with effective resource utilization.

3) NETWORK RESILIENCY

This is realized by incorporating redundancy, failover mechanisms, and active fault detection, which together enable flexibility and robustness.

4) SECURITY

Robust DDoS protection, encryption of traffic, and Zero Trust security modeling have been implemented to secure the infrastructure from security assaults.

5) COST EFFICIENCY

Direct peering and virtualization reduce third-party transit services expenses. In equipment and infrastructure can be a barrier to entry for smaller stakeholders, although the long-term savings will offset these costs.

6) HIGH UP-FRONT INVESTMENT

In equipment and infrastructure can be a barrier to entry for smaller stakeholders, although the long-term savings will offset these costs.

7) INTEROPERABILITY

Legacy systems can necessitate major upgrades to accommodate the new IXP infrastructure.

8) SECURITY THREATS

As 5G connectivity continues to unfold, it can be expected that new vulnerabilities will manifest, requiring the ongoing evolution of security measures.

9) ENVIRONMENTAL IMPACT

Heavy energy use, along with cooling needs, is compelling IXPs towards sustainable solutions, including renewable energy sources in combination with cooling mechanisms that support energy efficiency.

IX. FUTURE WORK

This section outlines a realistic deployment plan for the envisioned Internet Exchange Point (IXP) infrastructure. The

deployment is structured in several phases, beginning with controlled pilot trials over a restricted environment, e.g., a regional Internet Exchange Point or an academic testbed. In this way, latency, throughput, scalability, and security performance can be compared in real-world scenarios. Following the successful completion of pilot trials, more extensive deployment can move forward in modular enhancement of existing infrastructures in a way that minimizes operational disruption. Cost considerations are also addressed. While the upfront capital investment is high with the deployment of SDN and NFV technologies, long-term benefit is lower operational expense due to automation, improved traffic optimization, and less reliance on proprietary hardware. Financing alternatives, including institutional partnerships and grants, may be sought after in the initial stages of deployment. Actual on-site testing in the field should give insights regarding economic viability, longevity of performance, and regulatory adjustments to be made prior to widespread implementation.

X. CONCLUSION

The architecture presented for Internet Exchange Points (IXPs) designed for the routing of data related to 5G adopts new features with innovative ideas and requirements for routing through next-generation networks. The rapid development of 5G technology, which is expected to change many industries with enhanced connectivity, low-latency applications, and a huge surge in the number of connected devices, calls for further development of the underlying infrastructure in order to handle these challenges sufficiently. This architecture meets these requirements by enabling low-latency routing, seamless scalability, high security, and optimal resource allocation. Other important components of the proposed IXP architecture are the principles of performance optimization, edge computing, and fast switching fabrics for the quick and efficient exchange of data. The design is modular, and SDN/NFV is introduced into the architecture in order to help ensure that the infrastructure of the IXP will support massive increases in data generated by applications under 5G and keep it flexible and reduce costs regarding operations.

Furthermore, the design of the architecture is very considerate of network resilience and security by ensuring that the components of the network implemented are equipped with advanced fault tolerance mechanisms, DDoS mitigation systems, and a Zero Trust model in safeguarding the network from possible threats. Direct peering and virtualization contribute to cost efficiency since there is a reduced reliance on third-party transit services, which is very important in ensuring long-term sustainability and competitiveness. Realization of such architecture brings with it its own set of challenges. The integration of SDN, edge computing, and the existing legacy systems is rather complicated; thus, careful planning and cooperation among all the involved stakeholders is mandatory, not to mention significant capital investment.

This transition may also bring with it the challenges of interoperability, operational adjustments, and dealing with the heightened security risks from a larger 5G attack surface. The architecture will also ensure that IXPs continue at the core of efficient, low-latency data exchange and continue to play a role in the innovation and growth of global internet infrastructure in an evolving digital landscape.

ACKNOWLEDGMENT

The authors would like to thank and extend their heartfelt gratitude to SIMAD University for their continuous support throughout this research project. Their provision of resources and encouragement has been instrumental in the successful completion of this work.

REFERENCES

- [1] S. C. Gupta, G. Gupta, and H. Saran, "New vision for 5G backbone network architecture," in *Proc. IEEE 3rd 5G World Forum (5GWF)*, Sep. 2020, pp. 330–336.
- [2] D. Wagner, M. Wichtlhuber, C. Dietzel, J. Blendin, and A. Feldmann, "P4IX: A concept for P4 programmable data planes at IXPs," in *Proc. ACM SIGCOMM Workshop Future Internet Routing Addressing*, 2022, pp. 72–78, doi: [10.1145/3527974.3545725](https://doi.org/10.1145/3527974.3545725).
- [3] S. Chatzis, G. Smaragdakis, and A. Feldmann, "On the importance of Internet eXchange points for today's Internet ecosystem," Jul. 2013, *arXiv:1307.5264*.
- [4] K. W. Hon, "Networking/Internet infrastructure and technologies," in *Technology and Security for Lawyers and Other Professionals*. Northampton, MA, USA: Edward Elgar, 2024, pp. 288–316.
- [5] V. Vomhoff, M. Sichermann, S. Giebler, A. Lutu, M. Giess, and T. Hübfield, "A shortcut through the IPX: Measuring latencies in global mobile roaming with regional breakouts," in *Proc. 8th Netw. Traffic Meas. Anal. Conf. (TMA)*, May 2024, pp. 1–10.
- [6] S. Mirtchev and M. Mirtchev, "IP network interconnection—survey," *Electrotechnica Electronica*, vol. 55, pp. 1–11, May 2020.
- [7] T. Hoeschele, C. Dietzel, D. Kopp, F. H. P. Fitzek, and M. Reisslein, "Importance of internet exchange point (IXP) infrastructure for 5G: Estimating the impact of 5G use cases," *Telecommun. Policy*, vol. 45, no. 2, 2021, Art. no. 102091, doi: [10.1016/j.telpol.2020.102091](https://doi.org/10.1016/j.telpol.2020.102091).
- [8] A. Basit, S. Qaisar, M. Ali, M. Naeem, M. Bruyere, and J. J. P. C. Rodrigues, "Interconnecting networks with optimized service provisioning," *Telecommun. Syst.*, vol. 73, no. 2, pp. 223–239, Feb. 2020.
- [9] O. Gonzalez de Dios, R. Casellas, F. Cugini, and J. Alberto Hernandez, "Beyond 5G domainless network operation enabled by multiband: Toward optical continuum architectures," 2023, *arXiv:2302.08244*.
- [10] G. Biczok, M. Dramitinos, L. Toka, P. E. Heegaard, and H. Lonsethagen, "Manufactured by software: SDN-enabled multi-operator composite services with the 5G exchange," *IEEE Commun. Mag.*, vol. 55, no. 4, pp. 80–86, Apr. 2017.
- [11] M. I. I. Alam, E. Anshelevich, K. Kar, and M. Yuksel, "Pricing for efficient traffic exchange at IXPs," *IEEE/ACM Trans. Netw.*, vol. 31, no. 2, pp. 1234–1248, Feb. 2023, doi: [10.1109/TNET.2023.3336352](https://doi.org/10.1109/TNET.2023.3336352).
- [12] A. Agrawal and V. Bhatia, "Future backbone optical networks: Fiber densification versus network densification," in *Proc. IEEE Int. Conf. Adv. Netw. Telecommun. Syst. (ANTS)*, Dec. 2021, pp. 390–395.
- [13] J. Pan, J. Zhao, and L. Cai, "Measuring a low-earth-orbit satellite network," in *Proc. IEEE 34th Annu. Int. Symp. Pers., Indoor Mobile Radio Commun. (PIMRC)*, Sep. 2023, pp. 1–6.
- [14] D. Ponce, C. Tipantuña, and C. Espinosa, "Analysis of Internet traffic in Ecuador," *IEEE Access*, vol. 11, pp. 126365–126385, 2023.
- [15] J. Wang, M. Bewong, and L. Zheng, "SD-WAN: Hybrid edge cloud network between multi-site SDDC," *Comput. Netw.*, vol. 250, Aug. 2024, Art. no. 110509.
- [16] D. O'Briain, D. Denieffe, D. Okello, and Y. Kavanagh, "The Internet in east Africa, a mixed methods study," *East Afr. J. Sci., Technol. Innov.*, vol. 2, no. 1, pp. 1–15, Dec. 2020.
- [17] K. Tolga Bagci and A. Murat Tekalp, "SDN-enabled distributed open exchange: Dynamic QoS-path optimization in multi-operator services," *Comput. Netw.*, vol. 162, Oct. 2019, Art. no. 106845.
- [18] S. Greenstein, "The basic economics of Internet infrastructure," *J. Econ. Perspect.*, vol. 34, no. 2, pp. 192–214, May 2020.
- [19] N. Rahmatov and H. Baek, "Exploring scale-free networks: Survey on autonomous system dynamics and connectivity analysis," *Comput. Netw.*, vol. 248, Jun. 2024, Art. no. 110454.
- [20] K. Bilal, O. Khalid, A. Erbad, and S. U. Khan, "Potentials, trends, and prospects in edge technologies: Fog, cloudlet, mobile edge, and micro data centers," *Comput. Netw.*, vol. 130, pp. 94–120, Jan. 2018.
- [21] V. Stocker, G. Smaragdakis, and W. Lehr, "The state of network neutrality regulation," *ACM SIGCOMM Comput. Commun. Rev.*, vol. 50, no. 1, pp. 45–59, Mar. 2020.



OSMAN DIRIYE HUSSEIN is currently a Lecturer with SIMAD University, Mogadishu, Somalia. His academic and research interests focus on communication systems, smart technologies, and the design and optimization of next-generation network infrastructures. He has authored more than nine Scopus-indexed publications in areas including the Internet of Things (IoT), internet exchange points (IXPs), the IoT in blockchain environments, and artificial intelligence (AI) in telecommunication systems. His work emphasizes enhancing connectivity, optimizing network performance, and promoting digital inclusion across emerging regions.



ADAM MUHUDIN is currently a Lecturer with SIMAD University, specializing in information security. His research interests include reversible data hiding in encrypted domains and the development of secure system architectures. He explores advanced cryptographic techniques to enhance data confidentiality and integrity. His work also addresses security challenges in emerging technologies, such as the Internet of Things (IoT), blockchain, and cloud robotics. He investigates methods to secure interconnected devices, decentralized systems, and cloud-based robotic frameworks. Committed to interdisciplinary approaches, he develops practical solutions that integrate cryptography with innovative technologies. He regularly publishes in international journals and conferences, contributing to advancements in cybersecurity and secure computing.

• • •